

# Hyperloop Development Program

Final Report

16-05-2025

V1.0

This document is the final bi-annual report for the grant project Hyperloop Development Program (HDP), under reference, 1300031782. The document will summarize the results of the HDP over the full subsidy period (2020-2024), and will specify the highlights the progress of the Hyperloop Development Program 2024-H2.

The first section summarizes the highlights within the hyperloop market in 2024-H2, the second section provides an overall program review, followed by a section of progress per working group, and summarized in the financial overview.

## **REPORTING ON HIGHLIGHTS WITHIN HYPERLOOP MARKET**

**The period 2024-H2 have shown highlights in the Global developments in the Hyperloop Market:**

July

- During the European Hyperloop Week in Zurich, Switzerland, university teams from around the world present prototype components and hyperloop pods to a panel of judges, competing in several categories.
- EuroTube inaugurates the first section of the DemoTube, which is a hyperloop test track that is made of concrete and includes a vacuum chamber with an airlock and valve.

August

- Hardt Hyperloop conducts the first vacuum test at the European Hyperloop Center, successfully reaching the target pressure of 1 millibar.

September

- Hardt Hyperloop performs a successful test with their test vehicle, moving through the first 90 meters of the test track at a speed of 30 km/h, at the European Hyperloop Center in Veendam, the Netherlands.
- TEKNOFEST, Türkiye's aerospace and technology festival, includes a hyperloop competition for magnetic levitation and suspension technologies.

October

- Hardt Hyperloop successfully tests the lane switch at the European Hyperloop Center. This technology is essential for building and operating future hyperloop transportation networks.

November

- Southwest Jiaotong University (SWJTU, Chengdu) showcases the first 300 meters of their facility which is planned to stretch over more than 1 km by summer 2025 - as shown in the picture on the right.

December

- The German Institute of Hyperloop Technology realizes the goTube testing infrastructure. The first tests are performed, which are aimed at testing propulsion systems and different vehicle concepts for freight.

- The Hyper4Rail project, supported by Europe's Rail, kicks off with a consortium of 27 partner organisations from 13 countries. The project, which is led by the HDP, aims to establish a common hyperloop design and create a roadmap for industrialization.
- Indian Railways, IIT Madras and TuTr Hyperloop achieve a milestone by completing the country's 410-meter hyperloop test track in Chennai, India.

## REPORTING ON OVERALL PROGRAM PROGRESS

The Program has been wrapped up by the end of 2024, with the final stage concluded over 2024-H2.

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period 2020-2024.

The key program deliverable and key deliverable for this stage per working group are represented below.

| WG                       | Key Program deliverable                                                                                | Key deliverable status (31 Dec 2024)                                             |
|--------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| WG Hyperloop Development | Demonstrated hyperloop for cargo system (TRL-7) in the EHC                                             | Vehicle tests have been performed at the EHC                                     |
| WG EHC                   | European Hyperloop Center (EHC) Phase A completed                                                      | EHC has been commissioned and handed-over to Stichting EHC.                      |
| WG FP                    | Identify and develop 5 routes up to a preliminary level, of which 1 with a go ahead build-out of route | Research Infrastructure Cluster established.<br>5 Routes have been assessed.     |
| WG Program Management    | Establish a connected hyperloop ecosystem                                                              | Hyper4Rail project, where HDP is the coordinator, has started per December 2024. |

A detailed overview of program deliverables for each of the activities within the working groups, linked with the research themes, is presented below. The table also includes an overview of the deliverables per activity. These deliverables are shared attached to this report.

| Task                         | ID   | Name                                                        | Safety | Standards | Integration | Adoption | Public Interest | Ecosystem | Technology | Final delivery Date | Deliverable Done                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|------|-------------------------------------------------------------|--------|-----------|-------------|----------|-----------------|-----------|------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Management</b>    |      |                                                             |        |           |             |          |                 |           |            |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| FP0.0                        | D1.0 | Progress Report WG                                          | x      | x         | x           | x        | x               | x         | x          | Q3 2023             | 1 HDP Reporting 2021-H1<br>2 HDP Reporting 2021-H2<br>3 HDP Reporting 2022-H1<br>4 HDP Reporting 2022-H2<br>5 HDP Reporting 2023-H1<br>6 HDP Reporting 2023-H2<br>7 HDP Reporting 2024-H1<br>8 HDP Reporting 2024-H2 & Final report                                                                                                                                                                                                 |
| <b>Hyperloop Development</b> |      |                                                             |        |           |             |          |                 |           |            |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| F1-C01                       | D1.1 | Delivery report on System Design                            |        |           |             |          |                 |           | x          | Q4 2023             | 1 [HDP-WG-CARGO-1.1] 2021-07-09 - WG Cargo - Conceptual Design Phase Summary - Signed<br>2 HDP-WG-Pax-1.14 (ADSE no 21-RA-012) Hyperloop Interior Design Guidelines v1.0<br>3 [HDP-WG-Pax-2.7] - Business Opportunity Definition - HDP<br>4 [HDP-WG-CARGO-2.15] 2022-02-08 EHCA - Preliminary Design Phase Report<br>5 07192022_Denys-Hardt Tunneling<br>6 CargoLoop Specification Overview<br>7 Hardt Hyperloop System Description |
| F1-P02                       | D2.2 | Delivery report on Safety                                   | x      |           |             |          |                 |           |            | Q4 2023             | 1 2021-04-02 - Safety Vision<br>2 SafetyConcept - Deliverable - 20210603<br>3 Swiss Tunnel Safety Concept                                                                                                                                                                                                                                                                                                                           |
| F1-P03                       | D2.4 | Delivery report on Public Adoption                          |        |           |             | x        |                 |           |            | Q4 2023             | 1 Delt. Hyperloop-FSR, Socioeconomic Aspects of Hyperloop Development (1)<br>2 2023-02-03 UNH-HH-DPP, Cargo Hyperloop<br>3 CargoLoop Operation Concept Description HDP<br>4 Market needs for Cargo hyperloop capabilities<br>5 Report User Perceptions of Hyperloop Thesis O Callahan                                                                                                                                               |
| F1-P04                       | D1.5 | Delivery Progress report on Standardization                 |        | x         |             | x        |                 |           |            | Q4 2023             | 1 [HDP-WG-CARGO-1.19] Certification RoadMap for the Hyperloop project of Hyperloop Development Program<br>2 Exec. Certification Roadmap, Final<br>3 D2.10 Hyperloop Standardization and Regulation Report - Signed                                                                                                                                                                                                                  |
| <b>EHC</b>                   |      |                                                             |        |           |             |          |                 |           |            |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R1-P01                       | D3.1 | Permits approved for EHC                                    |        |           |             |          |                 |           |            | Q2 2021             |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R1-P02                       | D3.2 | Build report of EHC                                         |        |           |             |          |                 |           | x          | Q2 2023             | 1 [HDP-WG-EHC-1.3] European Hyperloop Center Project Brief - V4.1<br>3 Initial Test Results: Levitation, Guidance & Propulsion Tests, Safety & Recovery Tests, Vacuum System Test                                                                                                                                                                                                                                                   |
| R1-P03                       | D3.3 | Document system test results EHC                            |        |           |             |          |                 |           |            | Q4 2023             |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| R1-P04                       | D3.4 | Project closure document                                    |        |           |             |          |                 |           |            | Q4 2023             |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Future Prospects</b>      |      |                                                             |        |           |             |          |                 |           |            |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| FP1.1                        | D4.1 | Delivery report on global market analysis                   |        |           |             |          |                 |           | x          | Q3 2021             | Cancelled and we also don't report any costs under this WP                                                                                                                                                                                                                                                                                                                                                                          |
| FP1.2                        | D4.2 | Delivery report on preliminary route studies                |        |           |             |          |                 | x         | x          | Q3 2023             | 1 Route Study #1: [HDP-WG-FP-3.4] - Hyperconnected-Europe-Phase-1 Report<br>2 Route Study #2: Kuala Lumpur - Singapore<br>3 Route Study #3 Delhi - Ahmedabad<br>4 Route Study #4 Hannover - Bräunfeld<br>5 Route Study #5 Brisbane - Gold Coast<br>6 Impact Hyperloop op de ruimtelijke ordening 2050<br>7 [HDP-WG-FP-2.6] Cargo-Hyperloop-Holland, Full Report                                                                     |
| FP1.3                        | D4.3 | Report on need for future hyperloop research infrastructure |        |           |             |          |                 | x         | x          | Q1 2023             | 1 [HDP-WG-FP-1.18] WP 5.1 Inventory of Already Available Testing Environments<br>2 Delt. Hyperloop-FSR, Technical Aspects of Hyperloop Systems (1)<br>3 Future hyperloop research infrastructure                                                                                                                                                                                                                                    |
| FP1.4                        | D4.4 | Report on framework for executing hyperloop competitions    |        |           |             |          |                 |           | x          | Q4 2021             | 1 [HDP-WG-FP-1.13] - Framework for Hyperloop Competitions 1.2                                                                                                                                                                                                                                                                                                                                                                       |
| FP1.5                        | D4.5 | Report on hyperloop roadmap and outlook                     |        |           |             |          |                 |           | x          | Q2 2023             | 1 [HDP-WG-FP-1.12] - Outlook   Speeding up swiftly<br>2 Outlook II - Perspectives for hyperloop developments in NL                                                                                                                                                                                                                                                                                                                  |

## REPORTING PER WORKING GROUP

### WG Hyperloop Development and WG EHC

The report for working Group Hyperloop Development and EHC have been combined, with the main results the commissioning of the European Hyperloop Center, and the Vehicle tests at the EHC. In August the vehicle was transported to the EHC for the first time, to perform the vehicle tests, at the same time, the vacuum system tests have been performed to determine the low pressure hold of the hyperloop infrastructure.

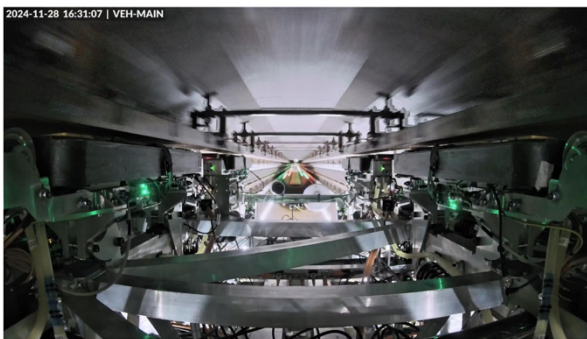
### **Vehicle Tests**

In 2024-H2, the following critical tests have been performed with the vehicle at the EHC.

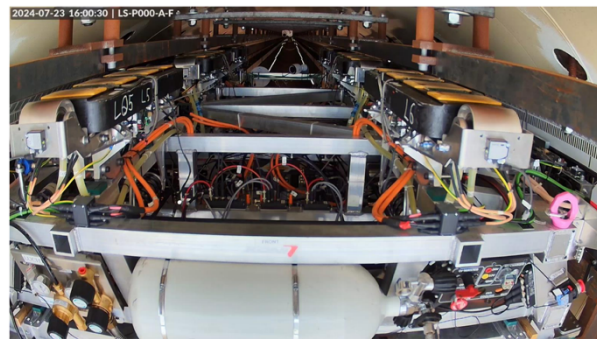
- Stable levitation & Guidance
- Propulsion in linear section
- Traversal of high-speed leg switch
- Traversal of low-speed leg switch



On 24 July 2024, a successful run was completed where the vehicle reached stable levitation and guidance in the EHC infrastructure for >5 minutes whilst also performing low-speed propulsion in the first tube.



Vehicle Switch test



On 28 November, a successful run was performed at 6 m/s through the high-speed leg of the switch. This critical test demonstrated the ability of the hyperloop vehicle to switch tubes without moving parts on the infrastructure. As such, in the coming period the speed of the vehicle will gradually increase.

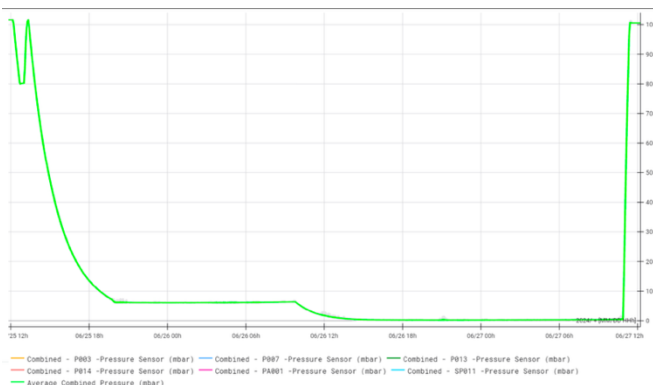


### Vacuum system test

On 26 June, the successful vacuum system test was performed. The objectives of the test were to verify structural strength of EHC infrastructure when under pressure load, to determine minimum pressure that can be achieved by the infrastructure + vacuum system (target: 1 mbar), and finally to determine the leak rate of the EHC infrastructure

Test Description. The Following steps were executed:

- Initial pumpdown to 800 mbar to load the infrastructure and check for initial (major) leaks.
- Repressurization to 1013 mbar to verify all steps in the testing procedure work as intended.
- Pumpdown till end of day (to ~60 mbar).
- Hold overnight for initial leak rate measurement @ 60mbar.
- Pumpdown till end of day, to reach 1 mbar.
- Repressurization to 1013 mbar.



### Safety & Recovery tests

During the test period, a number of safety & recovery tests have been performed. These tests focus on the interaction between the vehicle and the infrastructure in case the vehicle transitions into emergency state:

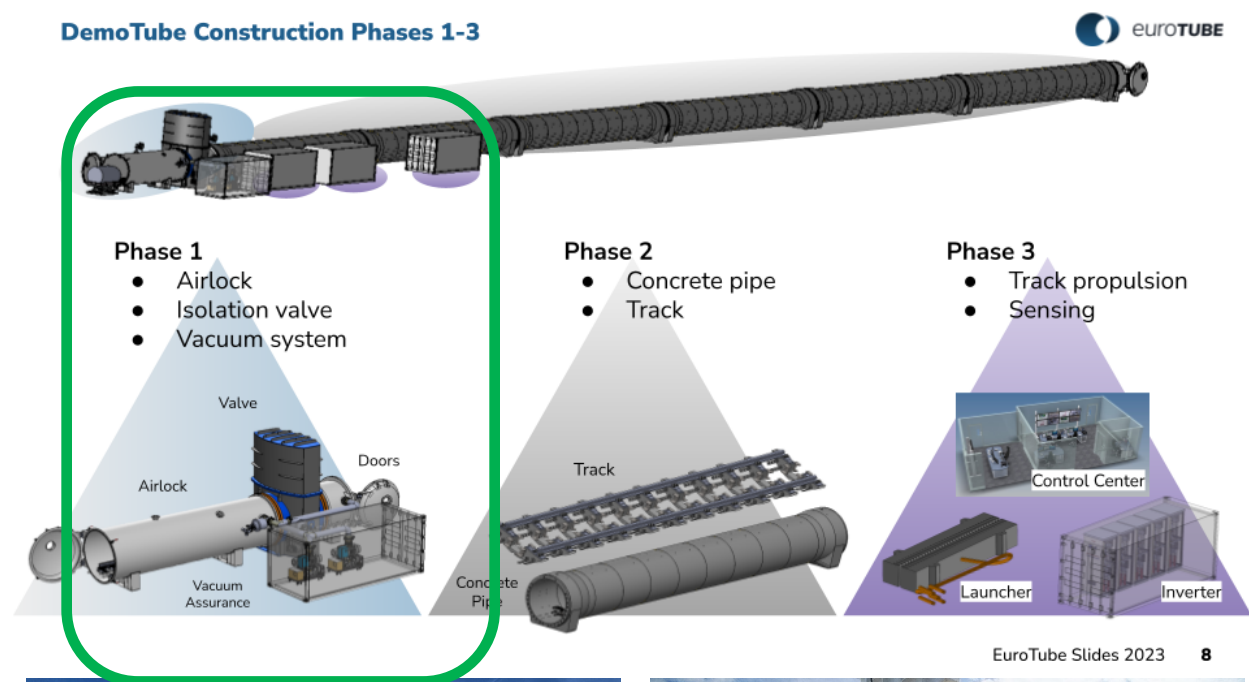
- Demonstration of vehicle Emergency Braking System (EBS).
- Demonstration of vehicle Catch System.
- Demonstration of vehicle recovery from infrastructure
- Demonstration of (safe) response in case of loss of vehicle power, loss of infrastructure power, or loss of communication

## EuroTube

In Q2 2024, EuroTube has kicked-off the construction of *DemoTube*, which is a 120m long demonstrator. As described further below, it's main purpose is the proof of concept of construction processes and the concept of sealed, modular post-tensioned concrete tubes. Since all construction steps are performed by EuroTube itself to maximize learning and measurement and data gain, construction will last until mid-2025.

In June, in time for the European Hyperloop Week (EHW), the Phase 1 of DemoTube was completed. This meant the assembly of the airlock, connecting it the BigValve and assuring its vacuum-tightness. Phase 1 also included the interior refurbishment of the Vacuum and control containers to fit the needs, piping, electrical and pneumatic works. Finally, the Control System was developed. With that, the main „active parts“ of DemoTube are completed and vacuum test within the airlock can be run.

The BigValve, the first valve suited for hyperloop application, was successfully tested. During EHW, the pod of Swissloop was tested in DemoTube for its leakage compensation mechanism.



## WG Future Prospects

### Global Market Assessment

This activity has been scrapped and the subsidy allocated for the activity has been redistributed to other activities.

### Preliminary Route Studies

In total seven preliminary route studies have been carried out, as summarized in the table below.

|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Cargo Hyperloop Holland study (Hyperconnected-Europe-Phase-1-Report) | This pre-feasibility study has been conducted to investigate the economic viability of the Cargo-hyperloop Holland corridor as a hyperloop network transporting cargo in the provinces of North Holland and South Holland. The aim for this study is threefold. Firstly, to advance a concept for hyperloop and ensure that the functionalities of the hyperloop system meet the needs and wishes of shippers in the region. Secondly, to design the corridor, including alignment, integration and hubs and perform a demand and socio-economic assessment. Finally, to identify and assess the feasibility of implementing a pilot route for hyperloop somewhere along this corridor and how to approach the project realization and governance.                                                                                                                                                                                                                                                                                                                                               |
| Kuala Lumpur -Singapore                                                  | This study, from April 2022, delves into the revolutionary transport technology of hyperloop and its potential to transform passenger experiences and cargo movements. Specifically, it examines the possibility of implementing a Kuala Lumpur to Singapore link using hyperloop as an alternative to high-speed rail plans that were recently suspended. Hyperloop has the ability to integrate with existing transport networks and contribute to city-shaping, place-making, economic growth, and sustainability, making it a potential game-changer for Malaysia, Singapore, and the broader Association of Southeast Asian Nations (ASEAN) economic community. This aligns with relevant megatrends in the region.                                                                                                                                                                                                                                                                                                                                                                         |
| Delhi - Ahmedabad                                                        | In this report the potential of using hyperloop technology to connect Ahmedabad and New Delhi in India is explored. The report includes projected passenger volumes, travel demand projections, and potential travel choice changes. It suggests that the 2, 4, and 14-hub options all have sufficient merit and recommends further project planning exercises through a collaborative partnership between Indian Railways and Hardt. Furthermore, the report recommends considering hyperloop as an intervention case by 2031 and as an alternative to high-speed rail for future high-speed corridors across India.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Hannover - Bielefeld                                                     | This study is a collaboration between Deutsche Bahn Engineering & Consulting and Hardt Hyperloop to compare the potential of hyperloop with high-speed rail (HSR) and maglev systems to upgrade the Hanover-Bielefeld rail corridor in Germany. The study aims to determine whether hyperloop has strategic merit as an alternative to the current upgrade options, considering factors such as reducing travel times, expanding commutable access to labor markets, developing a zero-emissions concept, generating new jobs, and developing a scalable concept of operations. The study includes the development of a hyperloop corridor between Hanover and Bielefeld, specification of optimal hyperloop fleet/vehicle type and configuration for passenger and cargo operations, and specification of a hyperloop service/operational concept for the fleet and hubs along the corridor. The study also presents three possible route alignments for the hyperloop corridor, with proposed hubs for passenger and cargo in locations along the corridor and across a full-network scenario. |
| Brisbane - Gold Coast                                                    | This report explored the potential of a direct hyperloop shuttle between Brisbane and the Gold Coast in Southeast Queensland. With the emergence of new modes of transportation, strategic planners are advised to consider the potential opportunities these new modes can bring. The proposed hyperloop scenario aims to address population growth, road congestion, and rail uncompetitiveness by providing a faster and more efficient mode of transportation. This report discusses routes, estimated demand for hyperloop services and the potential (financial) benefits are discussed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Impact Hyperloop op de ruimtelijke ordening 2050                         | This report is a synthesis of the presentations, discussions and input during a series of workshops "Impact hyperloop op Nederlandse Ruimtelijke Ordening 2050", held on 23 September 2022, 9 December 2022, and 9 May 2023, organized by the Hyperloop Development Program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| HyperConnected                                                           | A hyperloop network covering close to 25,000 kilometers would connect all European Union member states (except for islands), member candidates, and adjacent economies. Building this network would require an estimated investment of €981 billion, or about 10 percent of the total transport infrastructure investment required for Europe up to the year 2050. By this year, an estimated 1.3 billion passengers (1.3 trillion passenger-kilometers) and 625 million tonnes of freight (389 billion tonne-kilometers) could be transported by hyperloop, thereby reducing flight kilometers travelled by 66                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|  |                                                                                                                                                                                                                                                                                                                   |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | percent and long-haul truck movements by 18 percent. This could result in a reduction of between 113 and 242 million tonnes of CO2 per annum by 2050. The costs of hyperloop investment could be recovered through passenger fares and transport rates similar to aviation and trucking, reducing the total cost. |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### The HDP testing and research infrastructure cluster

In order to maximize the knowledge gained from these infrastructures, HDP has, together with EuroTube, Hochschule Emden-Leer, and EHC, started the “HDP testing infrastructure cluster”, to align European Test facilities into a collaborative cluster. As part of this Infrastructure Cluster, the partners have been working on a Test & Demonstration plan for passenger test facility.

The objective of this phase is to achieve a harmonized preliminary system integration specification by utilizing existing and proposed European hyperloop R&D facilities. This stage focuses on demonstrating and de-risking the core operational aspects of hyperloop and aligning technology development with future standards. This phase is i.a. performed through the Hyper4Rail project (where HDP is the coordinator), in which final prototypes of key missing subsystems will be delivered, the overall concept harmonized, and the roadmap to industrialization created. The main testing facilities in Europe currently are (non-exhaustive list, refer to WP5.1 Inventory of Already existing testing facilities.):

- The existing European Hyperloop Center (EHC) in the Netherlands,
- The DemoTube facility in Switzerland (under construction)
- The realized TU Munich facility in Germany
- The proposed Zeleros facility in Spain
- Maglev test track by Nevomo
- GoTube by Institute of Hyperloop Technology

### Hyperloop Roadmap and Outlook

In November 2021 the first outlook, *Speeding up swiftly*, a first of the roadmap has been established. As a country specific development agenda for hyperloop, including high potential routes. It defines the roadmap for the Netherlands. This Dutch roadmap will be preceded and substantiated by two outlooks, of which the document in front of you is the first. The goal of this first outlook is to inform and inspire national and regional authorities and other key stakeholders regarding the potential of a hyperloop network in the Netherlands and Europe and the possibilities to contribute to the realization of the network. In this first outlook we will explore the current context in which the Hyperloop Development Program operates, combining different sources into one distinct story about hyperloop technology, the high-potential Dutch routes and the key stakeholders to realize the hyperloop ambition.

March 2023 a second outlook, *Crossroads: perspectives for hyperloop development in the Netherlands*, has been established, focused on Past and current pilot projects (using maglev technology or in other domains) and its lessons learned.

### Building International Relationships for Ecosystem Expansion

HDP coordinates the application for a large hyperloop concept harmonization project initiated by the European Commission under Europe's Rail Joint Undertaking (ERJU): AWP24-FA7: ROADMAP TOWARDS INDUSTRIALISATION AND HARMONIZED IMPLEMENTABLE CONCEPT. The project was started in December 2024 and will run for a period of two years.